Akash Technolabs

Python Streamlit

#PythonNotes

Streamlit

What is Streamlit?

- Streamlit is an open-source Python library that makes it easy to create and share beautiful, custom web apps for machine learning and data science.
- Streamlit is very user-friendly.

Why one should use Streamlit?

- Streamlit is the easiest way especially for people with no front-end knowledge to put their code into a web application:

  - No front-end (html, js, css) experience or knowledge is required.
  - You don't need to spend days or months to create a web app, you can create a really beautiful machine learning or data science app in only a few hours or even minutes.
  - It is compatible with the majority of Python libraries (e.g. pandas, matplotlib, seaborn).
  - Less code is needed to create amazing web apps.

Setting Up Your Environment

- First, you need to install Streamlit. It's as simple as running this command in your Python environment:

pip install streamlit

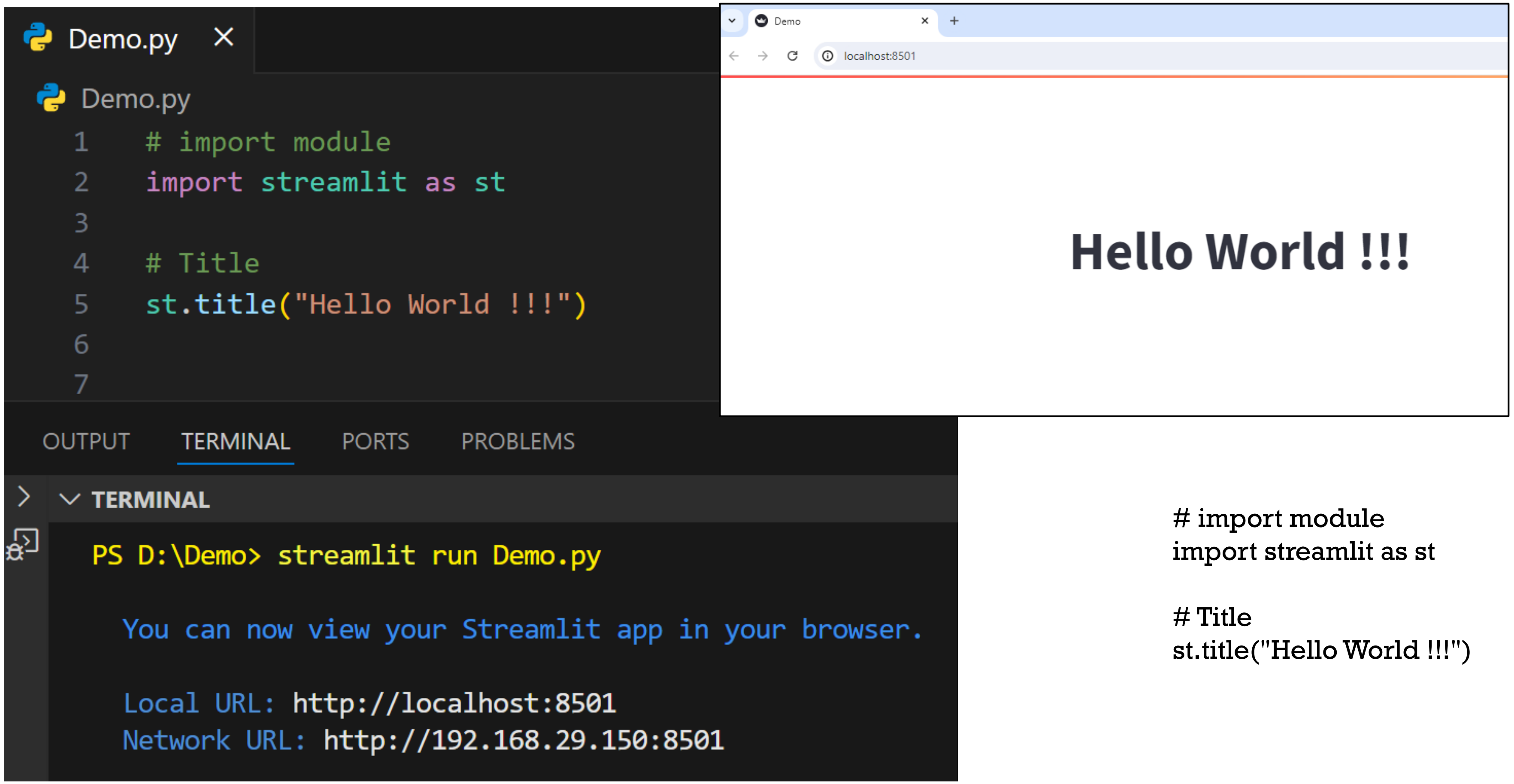
How to run Streamlit file?

- Open command prompt or Any type of terminal

streamlit run filename.py

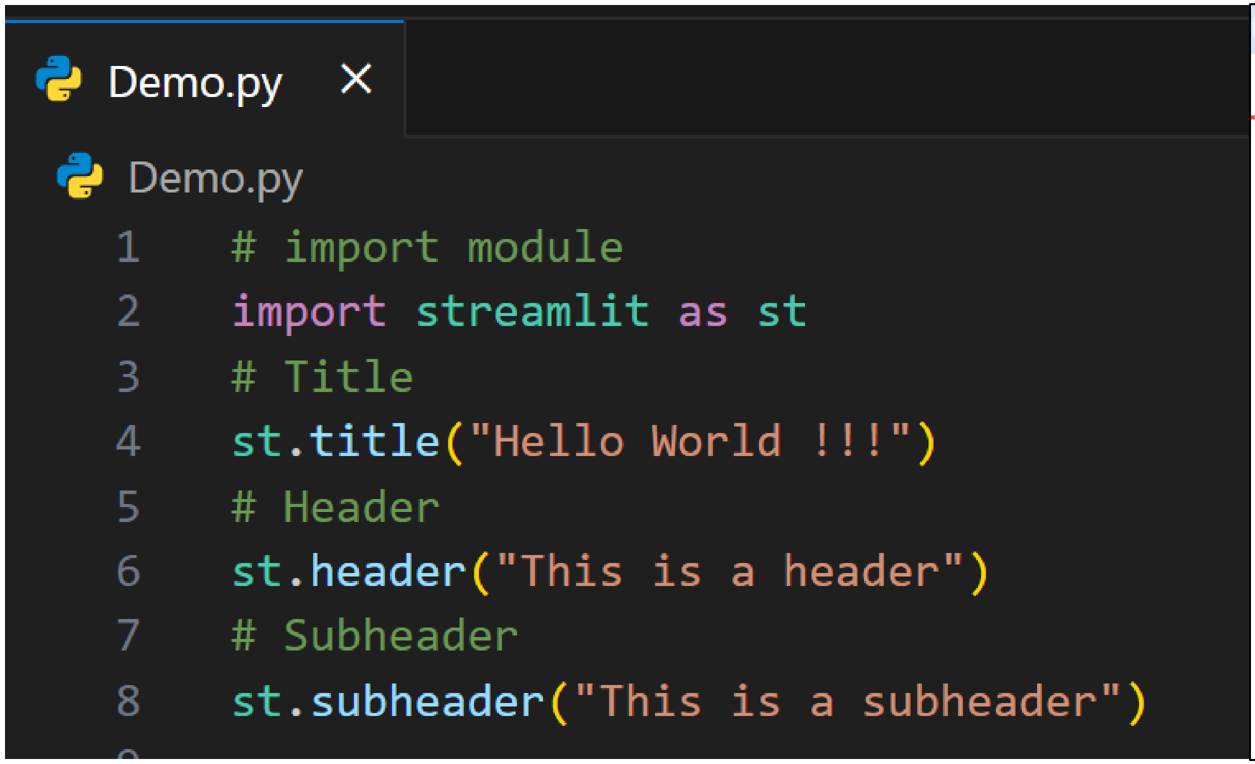
Streamlit basic functions

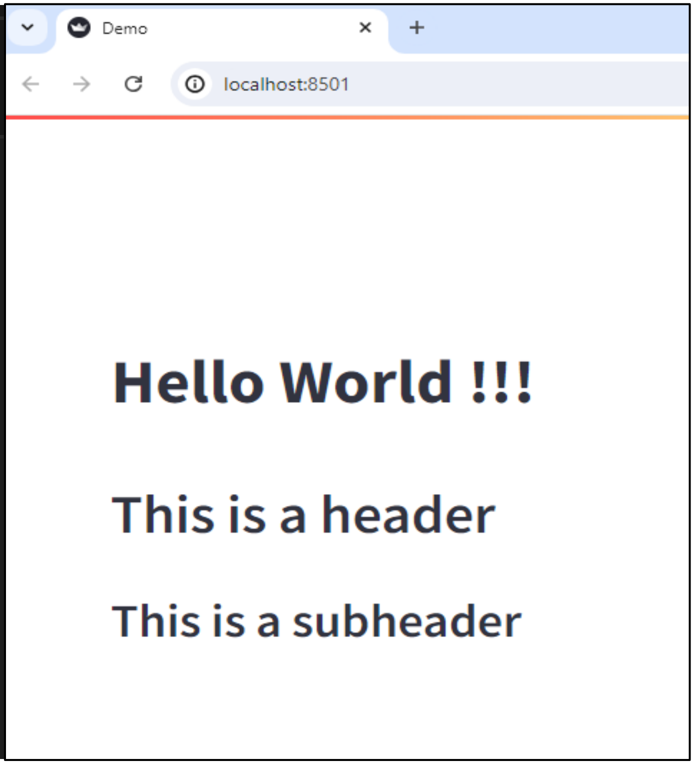
- Title:




```
# import module
import streamlit as st

# Title
st.title("Hello World !!!")
```

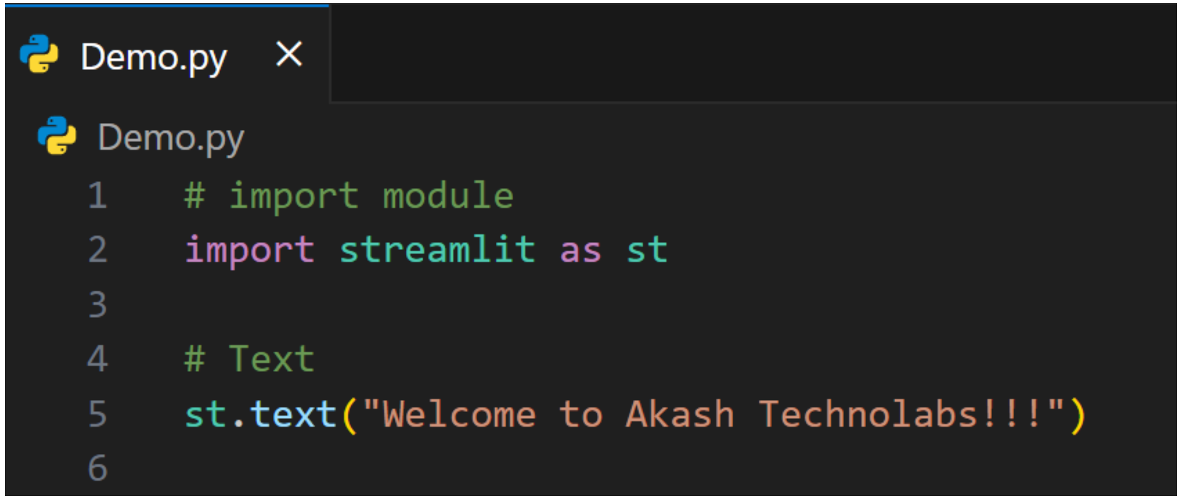
2. Header and Subheader:

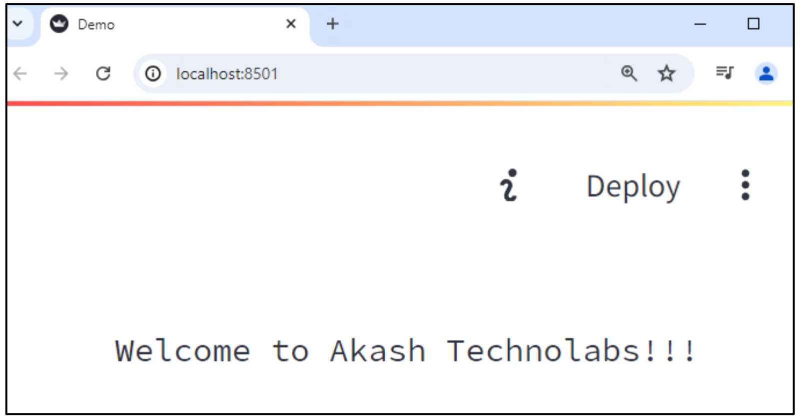





```
# import module
import streamlit as st
# Title
st.title("Hello World !!!")
# Header
st.header("This is a header")
# Subheader
st.subheader("This is a subheader")
```

3. Text:



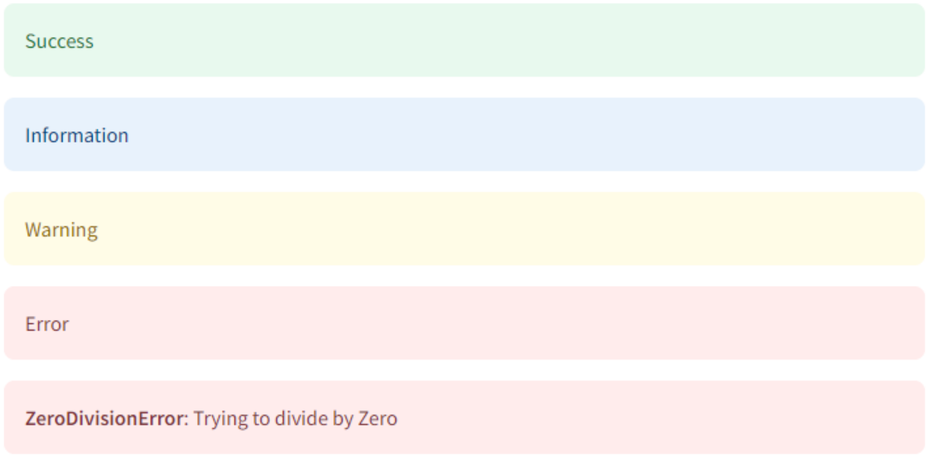



```
# import module
import streamlit as st

# Text
st.text("Welcome to Akash Technolabs!!!")
```

4. Success, Info, Warning, Error, Exception:

Demo.py

```python
# import module
import streamlit as st

# success
st.success("Success")

# success
st.info("Information")

# success
st.warning("Warning")

# success
st.error("Error")

# Exception - This has been added later
exp = ZeroDivisionError("Trying to divide by Zero")
st.exception(exp)
```




```python
# import module
import streamlit as st

# success
st.success("Success")

# success
st.info("Information")

# success
st.warning("Warning")

# success
st.error("Error")

# Exception - This has been added later
exp = ZeroDivisionError("Trying to divide by Zero")
st.exception(exp)
```

5. Write:

- Using write function, we can also display code in coding format. This is not possible using st.text(”).

Demo.py

```python
# import module
import streamlit as st

# Write text
st.write("Text with write")

# Writing python inbuilt function range()
st.write(range(10))
```

Text with write

`range(0, 10)`

# import module
import streamlit as st

# Write text
st.write("Text with write")

# Writing python inbuilt function range()
st.write(range(10))

6. Display Images:

Demo.py

Demo.py > ...

```python
# import module
import streamlit as st

# Display Images

# import Image from pillow to open images
from PIL import Image
img = Image.open("image1.jpg")

# display image using streamlit
# width is used to set the width of an image
st.image(img, width=200)
```

pip install pillow

# import module
import streamlit as st

Display Images

# import Image from pillow to open images
from PIL import Image
img = Image.open("image1.jpg")

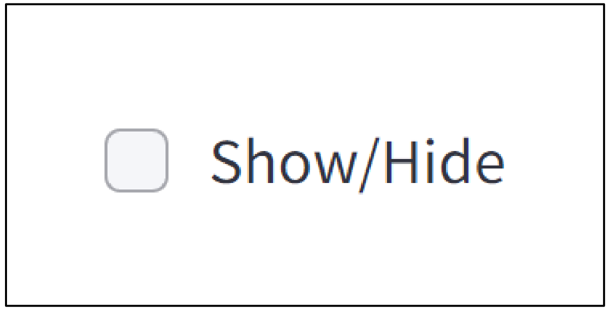
# display image using streamlit
# width is used to set the width of an image
st.image(img, width=200)

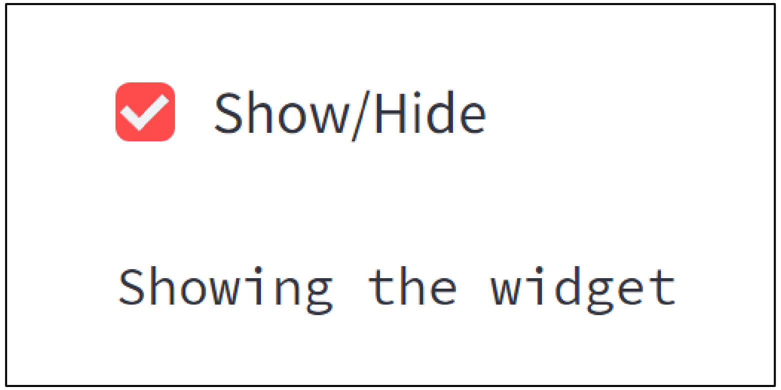
7. Checkbox:

- A checkbox returns a boolean value. When the box is checked, it returns a True value else returns a False value.

Demo.py

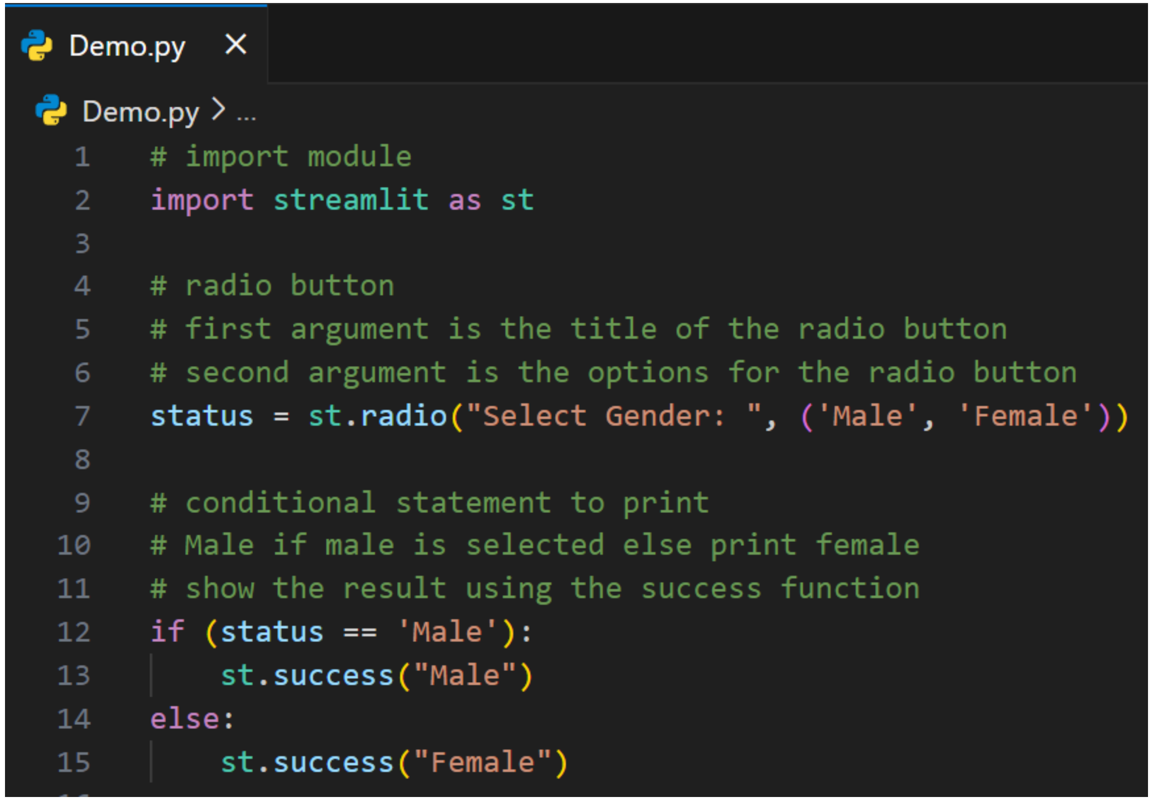
```python
# import module
import streamlit as st

# checkbox
# check if the checkbox is checked
# title of the checkbox is 'Show/Hide'
if st.checkbox("Show/Hide"):

    # display the text if the checkbox returns True value
    st.text("Showing the widget")
```

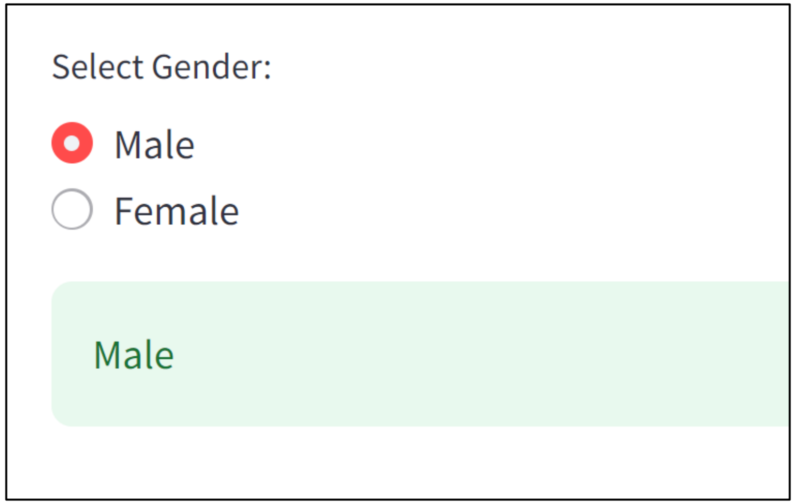





```python
# import module
import streamlit as st

# checkbox
# check if the checkbox is checked
# title of the checkbox is 'Show/Hide'
if st.checkbox("Show/Hide"):

        # display the text if the checkbox returns True value
        st.text("Showing the widget")
```

8. Radio Button:





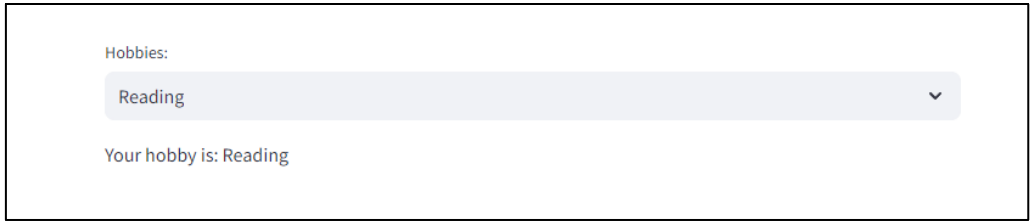

```
# import module
import streamlit as st

# radio button
# first argument is the title of the radio button
# second argument is the options for the radio button
status = st.radio("Select Gender: ", ('Male', 'Female'))

# conditional statement to print
# Male if male is selected else print female
# show the result using the success function
if (status == 'Male'):
        st.success("Male")
else:
        st.success("Female")
```

9. Selection Box:

- You can select any one option from the select box.

Demo.py

```python
# import module
import streamlit as st

# Selection box

# first argument takes the title of the selection box
# second argument takes options
hobby = st.selectbox("Hobbies: ",
                     ['Dancing', 'Reading', 'Sports'])

# print the selected hobby
st.write("Your hobby is: ", hobby)
```



# import module
import streamlit as st

Selection box

# first argument takes the title of the selection box
# second argument takes options
hobby = st.selectbox("Hobbies: ",
['Dancing', 'Reading', 'Sports'])

# print the selected hobby
st.write("Your hobby is: ", hobby)

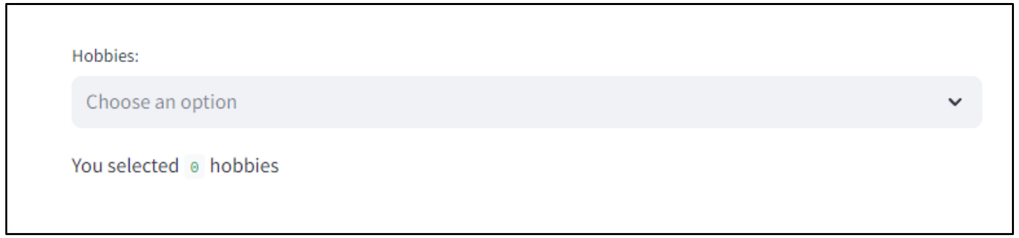
10. Multi-Selectbox:

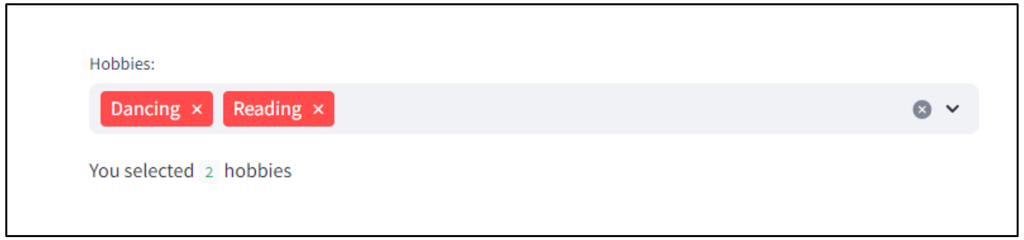
- The multi-select box returns the output in the form of a list. You can select multiple options.

Demo.py

Demo.py > ...

```python
# import module
import streamlit as st

# multi select box

# first argument takes the box title
# second argument takes the options to show
hobbies = st.multiselect("Hobbies: ",
                         ['Dancing', 'Reading', 'Sports'])

# write the selected options
st.write("You selected", len(hobbies), 'hobbies')
```






```python
# import module
import streamlit as st

# multi select box

# first argument takes the box title
# second argument takes the options to show
hobbies = st.multiselect("Hobbies: ",

                         ['Dancing', 'Reading', 'Sports'])

# write the selected options
st.write("You selected", len(hobbies), 'hobbies')
```

11. Button:

- st.button() returns a boolean value. It returns a True value when clicked else returns False.

Demo.py

```python
# import module
import streamlit as st

# Create a simple button that does nothing
st.button("Click me for no reason")

# Create a button, that when clicked, shows a text
if(st.button("About")):
    st.text("Welcome To Akash Technolabs...")
```

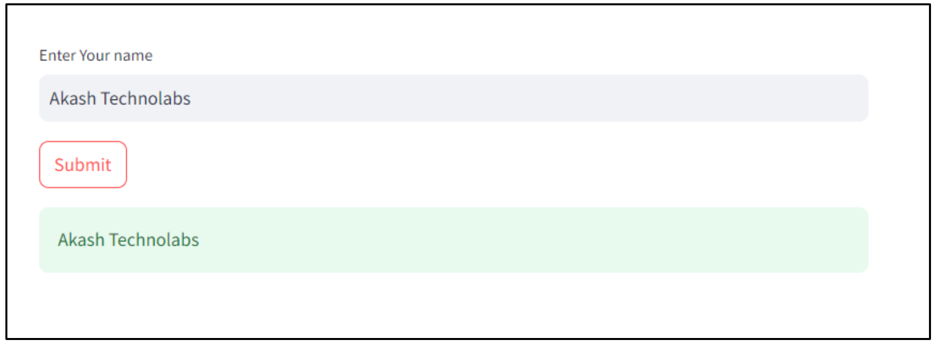

```python
# import module
import streamlit as st

# Create a simple button that does nothing
st.button("Click me for no reason")

# Create a button, that when clicked, shows a text
if(st.button("About")):
    st.text("Welcome To Akash Technolabs...")
```

12. Text Input:

Demo.py

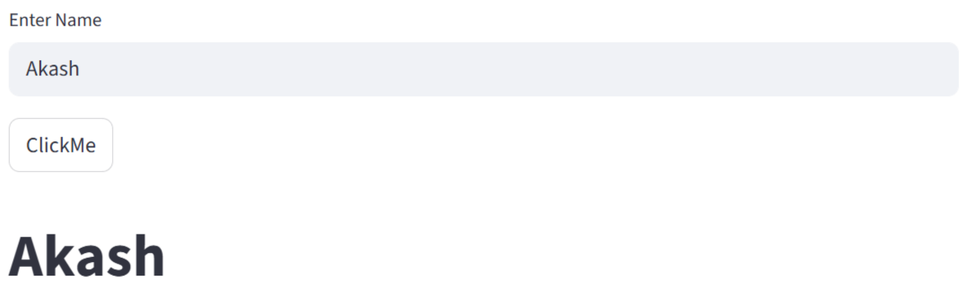
Demo.py > ...

```python
# import module
import streamlit as st

# Text Input
# save the input text in the variable 'name'
# first argument shows the title of the text input box
# second argument displays a default text inside the text input area
name = st.text_input("Enter Your name", "Type Here ...")

# display the name when the submit button is clicked
# .title() is used to get the input text string
if(st.button('Submit')):
    result = name.title()
    st.success(result)
```

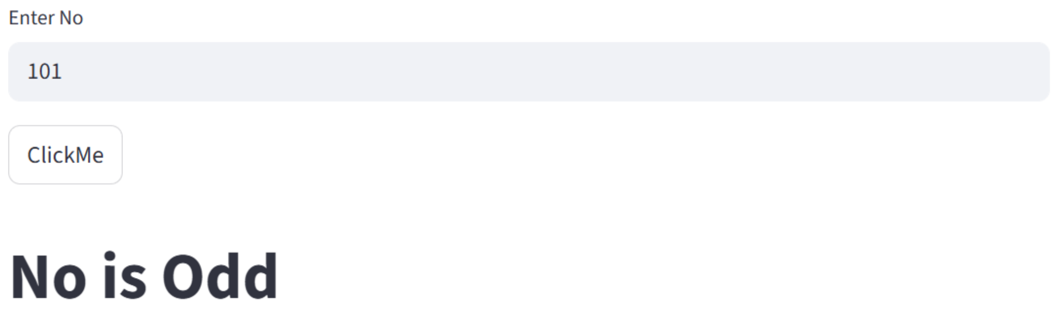



```python
# import module
import streamlit as st

# Text Input
# save the input text in the variable 'name'
# first argument shows the title of the text input box
# second argument displays a default text inside the text input area
name = st.text_input("Enter Your name", "Type Here ...")

# display the name when the submit button is clicked
# .title() is used to get the input text string
if(st.button('Submit')):
        result = name.title()
        st.success(result)
```

```python
import streamlit as st

name = st.text_input("Enter Name")
print(name)

if(st.button("ClickMe")):
    st.title(name)

```



```python
import streamlit as st

no1 = st.text_input("Enter No")

if(st.button("ClickMe")):
    if int(no1) % 2 == 0:
        st.title("NO is Even")
    else:
        st.title("No is Odd")
```

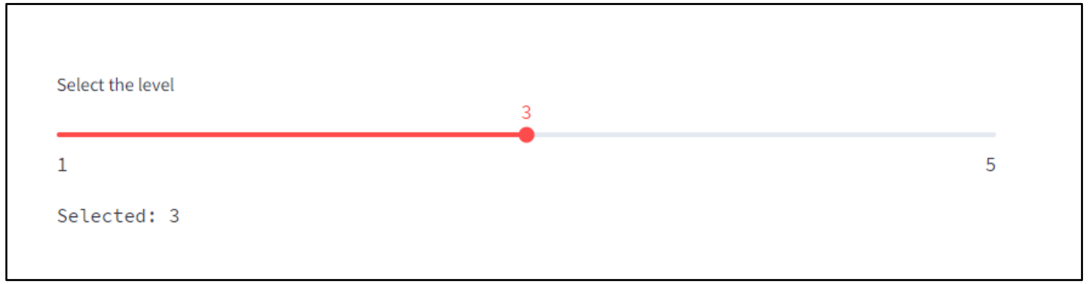


13. Slider:

Demo.py

Demo.py > ...

```python
# import module
import streamlit as st

# slider
# first argument takes the title of the slider
# second argument takes the starting of the slider
# last argument takes the end number
level = st.slider("Select the level", 1, 5)

# print the level
# format() is used to print value
# of a variable at a specific position
st.text('Selected: {}'.format(level))
```

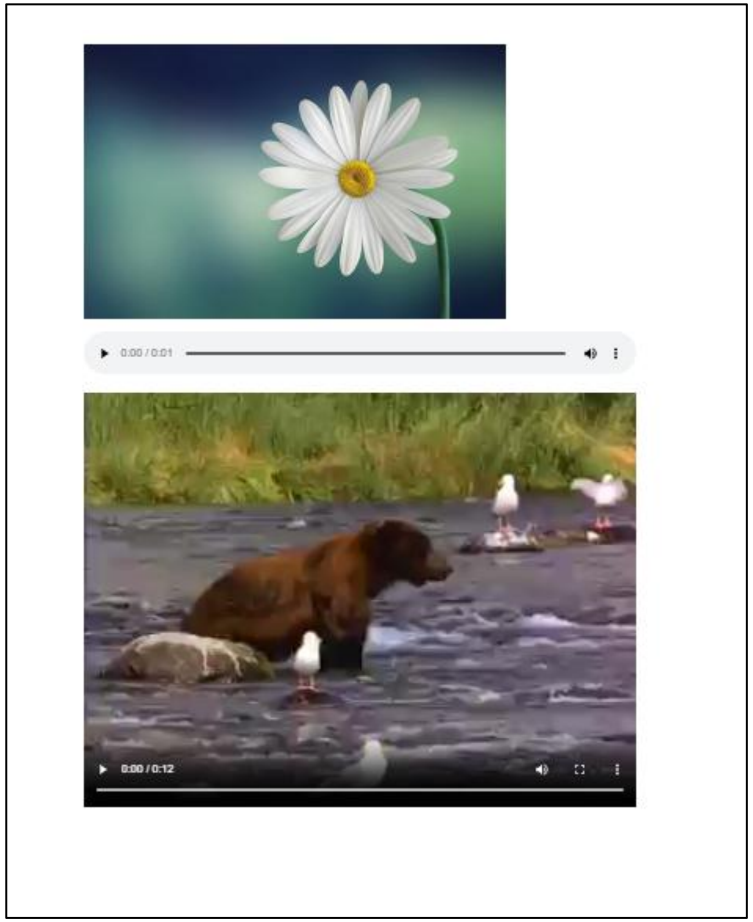



```python
# import module
import streamlit as st

# slider
# first argument takes the title of the slider
# second argument takes the starting of the slider
# last argument takes the end number
level = st.slider("Select the level", 1, 5)

# print the level
# format() is used to print value
# of a variable at a specific position
st.text('Selected: {}'.format(level))
```

14.Image/Audio/Video

Demo.py

```python
# import module
import streamlit as st

# Image
st.image("image1.jpg")

#Audio
st.audio("Audio.ogv")

#Video
st.video("Video.mp4")
```



# import module
import streamlit as st

# Image
st.image("image1.jpg")

#Audio
st.audio("Audio.ogv")

#Video
st.video("Video.mp4")

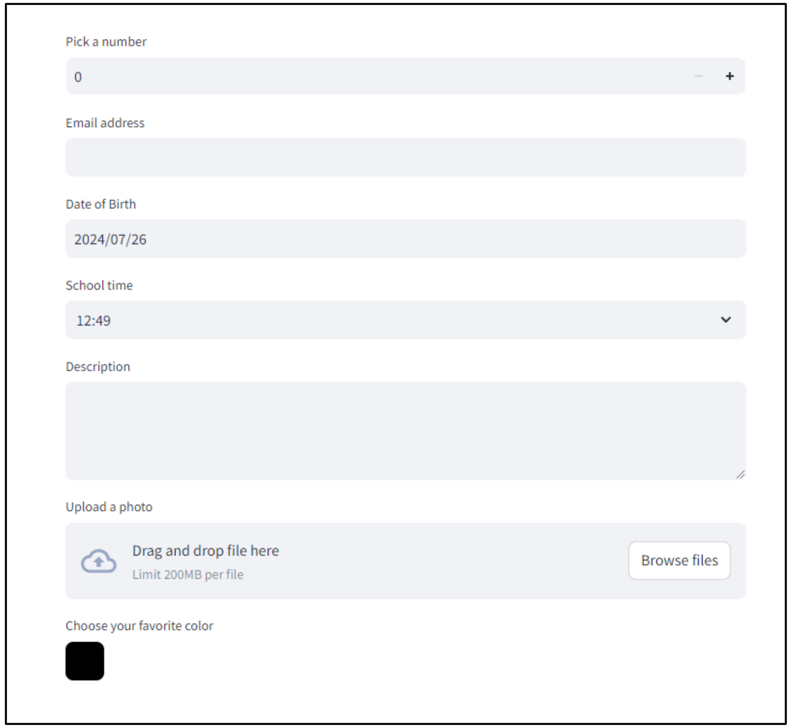
15:Input widgets

- st.number_input(): This function is used to display a numeric input widget.
- st.text_input(): This function is used to display a text input widget.
- st.date_input(): This function is used to display a date input widget to choose a date.
- st.time_input(): This function is used to display a time input widget to choose a time.
- st.text_area(): This function is used to display a text input widget with more than a line text.
- st.file_uploader(): This function is used to display a file uploader widget.
- st.color_picker(): This function is used to display color picker widget to choose a color.

Example

Demo.py

```python
# import module
import streamlit as st

st.number_input('Pick a number', 0,10)
st.text_input('Email address')
st.date_input('Date of Birth')
st.time_input('School time')
st.text_area('Description')
st.file_uploader('Upload a photo')
st.color_picker('Choose your favorite color')
```




```python
# import module
import streamlit as st

st.number_input('Pick a number', 0,10)
st.text_input('Email address')
st.date_input('Date of Birth')
st.time_input('School time')
st.text_area('Description')
st.file_uploader('Upload a photo')
st.color_picker('Choose your favorite color')
```

16:Display progress and status with Streamlit

- st.balloons(): This function is used to display balloons for celebration.
- st.progress(): This function is used to display a progress bar.
- st.spinner(): This function is used to display a temporary waiting message during execution.

Example

Demo.py

```python
# import module
import streamlit as st
import time

st.balloons()

st.subheader("Progress Bar")
st.progress(10)

st.subheader("Wait the execution")
with st.spinner('Wait for it...'):
    time.sleep(10)
```


```python
# import module
import streamlit as st
import time
st.balloons()
st.subheader("Progress Bar")
st.progress(10)
st.subheader("Wait the execution")
with st.spinner('Wait for it...'):
        time.sleep(10)
```

Sum of 2 Numbers

Demo.py

Demo.py > ...

```python
# import the streamlit library
import streamlit as st

# give a title to our app
st.title('Sum Of 2 Numbers')


n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
ans = n1 + n2

# check if the button is pressed or not
if (st.button('Calculate Sum')):

    st.text("Sum is  {}.".format(ans))

```

OUTPUT TERMINAL PORTS PROBLEMS

TERMINAL

```
PS D:\Demo> streamlit run Demo.py

  You can now view your Streamlit app in your browser.
```

Code

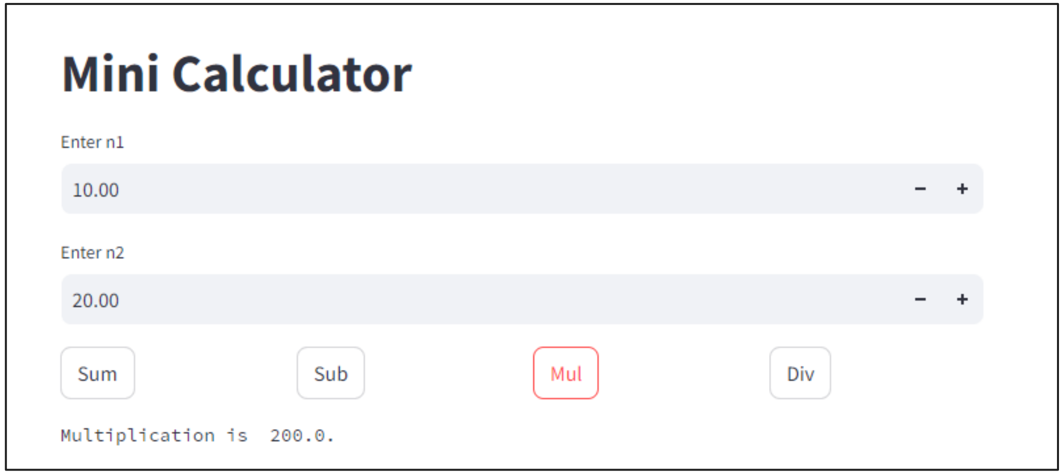
```
# import the streamlit library
import streamlit as st

# give a title to our app
st.title('Sum Of 2 Numbers')

# TAKE INPUT
n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
ans = n1 + n2

# check if the button is pressed or not
if (st.button('Calculate Sum')):
    # print the Sum
    st.text("Sum is  {}.".format(ans))
```

Mini Caculator

Demo.py

Demo.py > ...

```python
# import the streamlit library
import streamlit as st
# give a title to our app
st.title('Mini Calculator')
# TAKE  INPUT
n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
col1, col2, col3, col4 = st.columns(4)
with col1:
    Sum = st.button('Sum')
with col2:
    Sub = st.button('Sub')
with col3:
    Mul = st.button('Mul')
with col4:
    Div = st.button('Div')
if Sum:
    ans = n1 + n2
    st.text("Sum is  {}.".format(ans))
if Sub:
    ans = n1 - n2
    st.text("Subtraction is  {}.".format(ans))
if Mul:
    ans = n1 * n2
    st.text("Multiplication is  {}.".format(ans))
if Div:
    ans = n1 / n2
    st.text("Division is  {}.".format(ans))
```



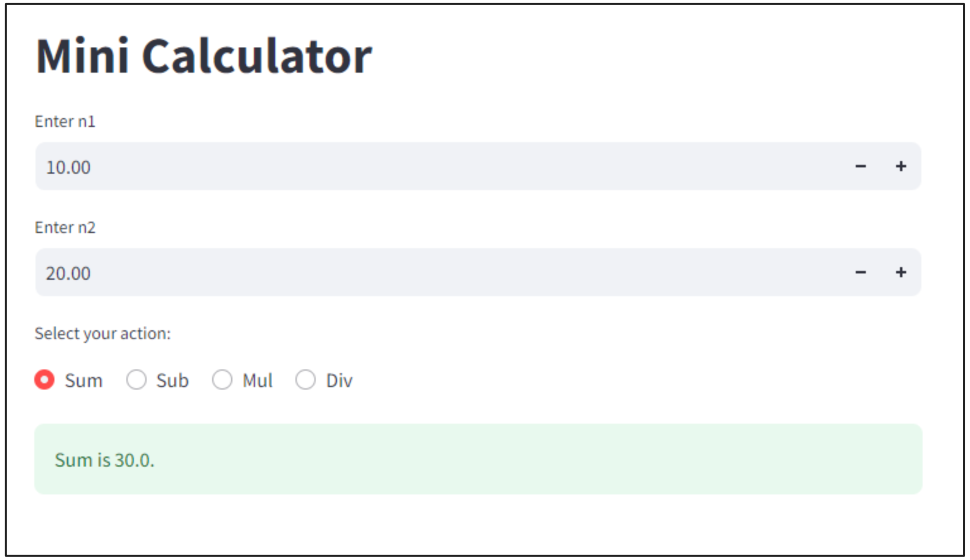
Code

```
# import the streamlit library
import streamlit as st
# give a title to our app
st.title('Mini Calculator')
# TAKE  INPUT
n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
col1, col2, col3, col4 = st.columns(4)
with col1:
   Sum = st.button('Sum')
with col2:
   Sub = st.button('Sub')
with col3:
   Mul = st.button('Mul')
with col4:
   Div = st.button('Div')
if Sum:
   ans = n1 + n2
   st.text("Sum is  {}.".format(ans))
if Sub:
   ans = n1 - n2
   st.text("Subtraction is  {}.".format(ans))
if Mul:
   ans = n1 * n2
   st.text("Multiplication is  {}.".format(ans))
if Div:
   ans = n1 / n2
   st.text("Division is  {}.".format(ans))
```

Mini Calculator Using Radio Button

Demo.py

```python
# import the streamlit library
import streamlit as st

# give a title to our app
st.title('Mini Calculator')
# TAKE  INPUT
n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
action = st.radio('Select your action: ',
                  ('Sum', 'Sub', 'Mul', 'Div'), horizontal=True)

if (action == 'Sum'):
    ans = n1 + n2
    st.success("Sum is  {}.".format(ans))
if (action == 'Sub'):
    ans = n1 - n2
    st.success("Subtraction is  {}.".format(ans))
if (action == 'Mul'):
    ans = n1 * n2
    st.success("Multiplication is  {}.".format(ans))
if (action == 'Div'):
    ans = n1 / n2
    st.success("Division is  {}.".format(ans))
```



Code

```
# import the streamlit library
import streamlit as st

# give a title to our app
st.title('Mini Calculator')
# TAKE  INPUT
n1 = st.number_input("Enter n1")
n2 = st.number_input("Enter n2")
action = st.radio('Select your action: ',
                  ('Sum', 'Sub', 'Mul', 'Div'), horizontal=True)

if (action == 'Sum'):
    ans = n1 + n2
    st.success("Sum is  {}.".format(ans))
if (action == 'Sub'):
    ans = n1 - n2
    st.success("Subtraction is  {}.".format(ans))
if (action == 'Mul'):
    ans = n1 * n2
    st.success("Multiplication is  {}.".format(ans))
if (action == 'Div'):
    ans = n1 / n2
    st.success("Division is  {}.".format(ans))
```

Sample Example

- https://currencyconverterra.streamlit.app/

- https://python-weather-app-ra.streamlit.app/

github.com/new

New repository

Type / to searc

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.

Required fields are marked with an asterisk (*).

1 **General**

Owner *

akashpadhiyar /

Repository name *

Great repository names are short and memorable. How about didactic-train?

Description

0 / 350 characters

2 **Configuration**

Choose visibility *

Choose who can see and commit to this repository

Public

Start with a template

Templates pre-configure your repository with files.

No template

Add README

READMEs can be used as longer descriptions. About READMEs

Off

Add .gitignore

.gitignore tells git which files not to track. About ignoring files

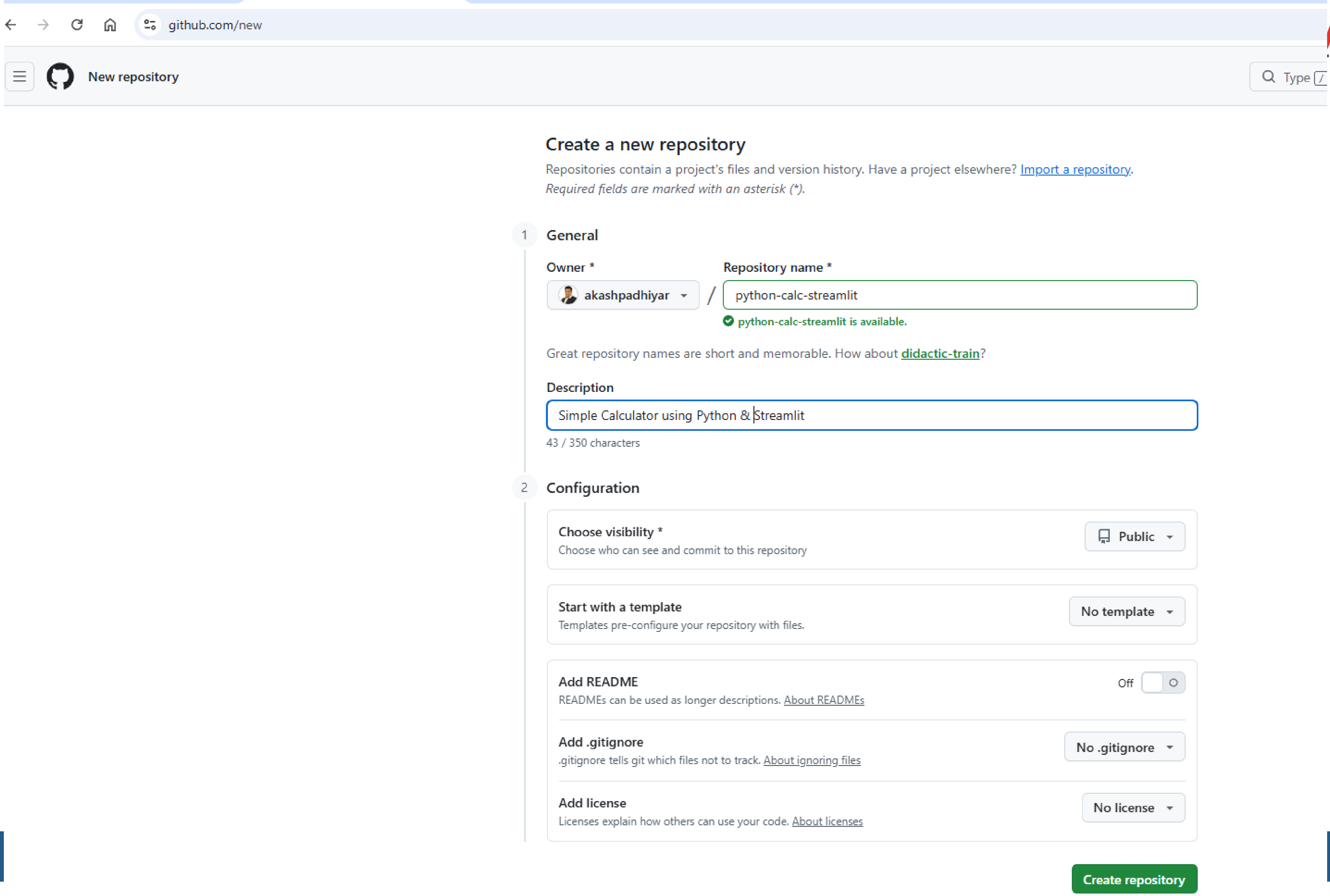
No .gitignore

Add license

Licenses explain how others can use your code. About licenses

No license

Create repository


github.com/new
New repository
Type /
Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1
General
Owner *
akashpadhiyar
/
Repository name *
python-calc-streamlit
python-calc-streamlit is available.
Great repository names are short and memorable. How about didactic-train?
Description
Simple Calculator using Python & Streamlit
43 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Start with a template
Templates pre-configure your repository with files.
No template
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository

github.com/akashpadhiyar/python-calc-streamlit

akashpadhiyar / python-calc-streamlit

Type / to search

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python-calc-streamlit Public

Pin | Watch 0 | Fork 0 | Star 0

Start coding with Codespaces

Add a README file and start coding in a secure, configurable, and dedicated development environment.

Create a codespace

Add collaborators to this repository

Search for people using their GitHub username or email address.

Invite collaborators

Quick setup — if you've done this kind of thing before

Set up in Desktop or HTTPS SSH https://github.com/akashpadhiyar/python-calc-streamlit.git

Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.

...or create a new repository on the command line

```
echo "# python-calc-streamlit" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/akashpadhiyar/python-calc-streamlit.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/akashpadhiyar/python-calc-streamlit.git
git branch -M main
git push -u origin main
```

Upload files · akashpadhiyar/py

github.com/akashpadhiyar/python-calc-streamlit/upload/main

akashpadhiyar / python-calc-streamlit

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Drag additional files here to add them to your repository

Or choose your files

streamlit_app.py

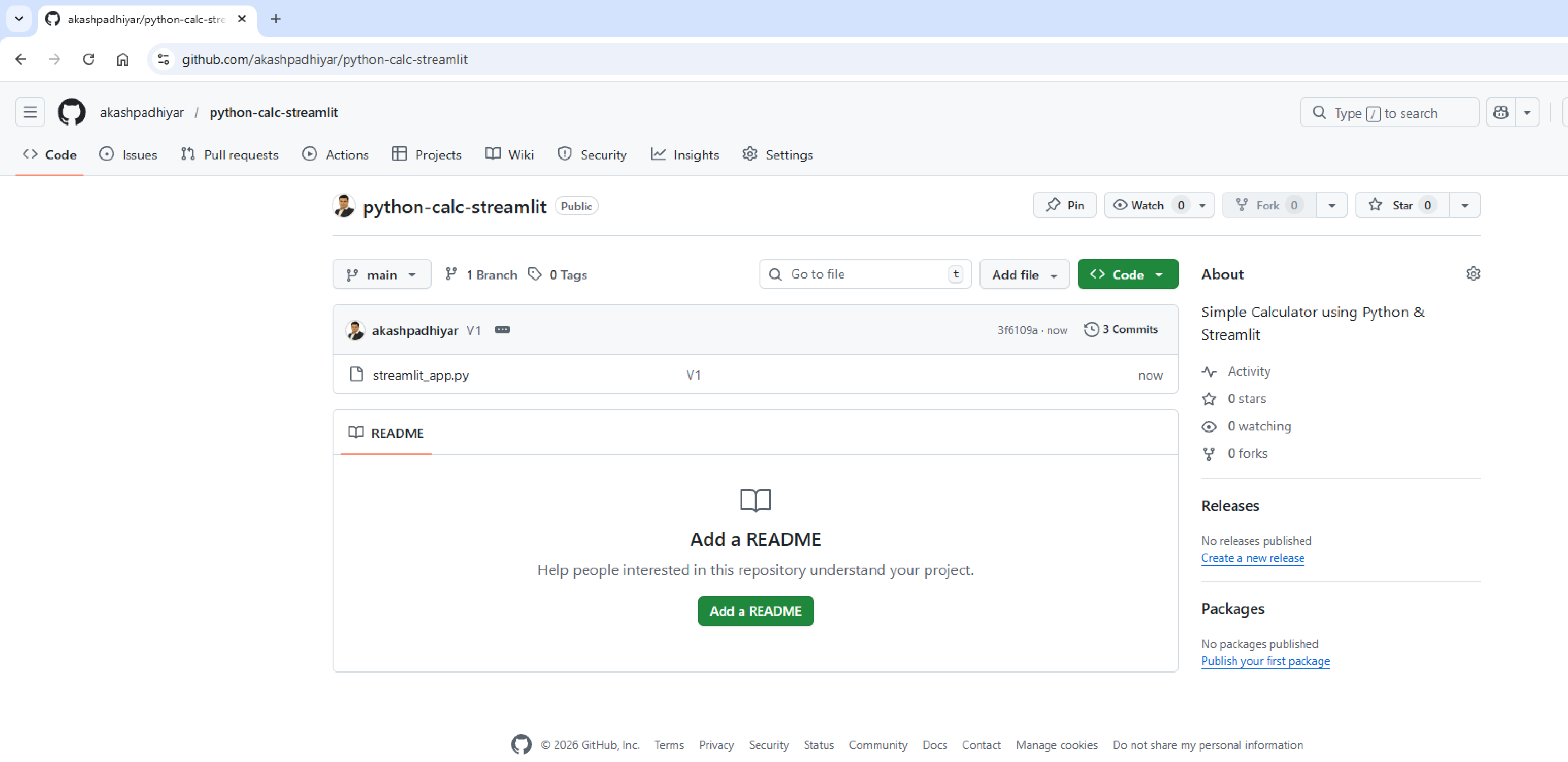
Commit changes

V1

V1

- Commit directly to the `main` branch.
- Create a **new branch** for this commit and start a pull request. Learn more about pull requests.

Commit changes Cancel


akashpadhiyar/python-calc-stre
github.com/akashpadhiyar/python-calc-streamlit
akashpadhiyar / python-calc-streamlit
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Settings
python-calc-streamlit
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
akashpadhiyar V1
3f6109a · now
3 Commits
streamlit_app.py
V1
now
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
Simple Calculator using Python & Streamlit
Activity
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0 watching
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Releases
No releases published
Create a new release
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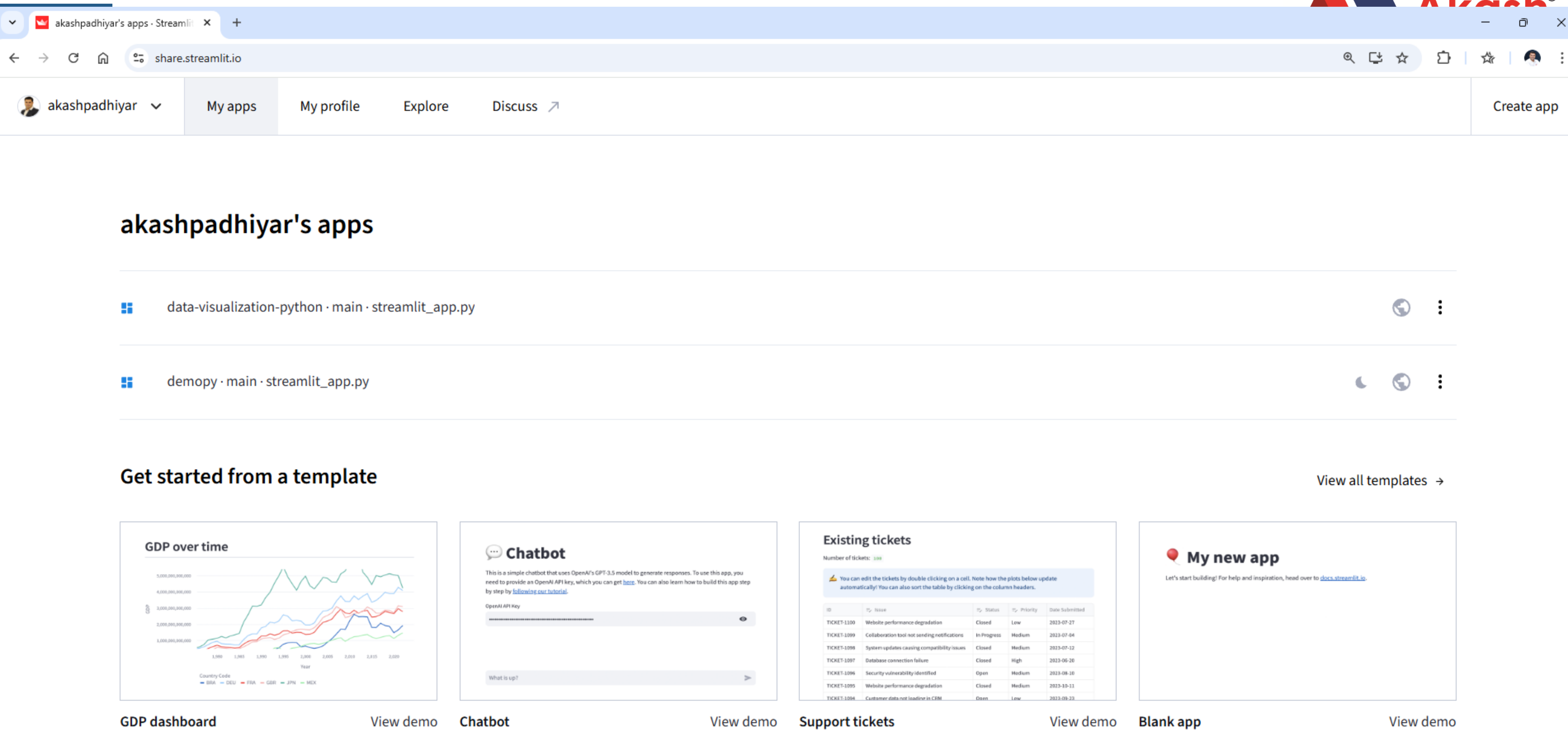
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Deploy an app

Repository
Paste GitHub URL

akashpadhiyar/python-calc-streamlit

Branch

main

Main file path

streamlit_app.py

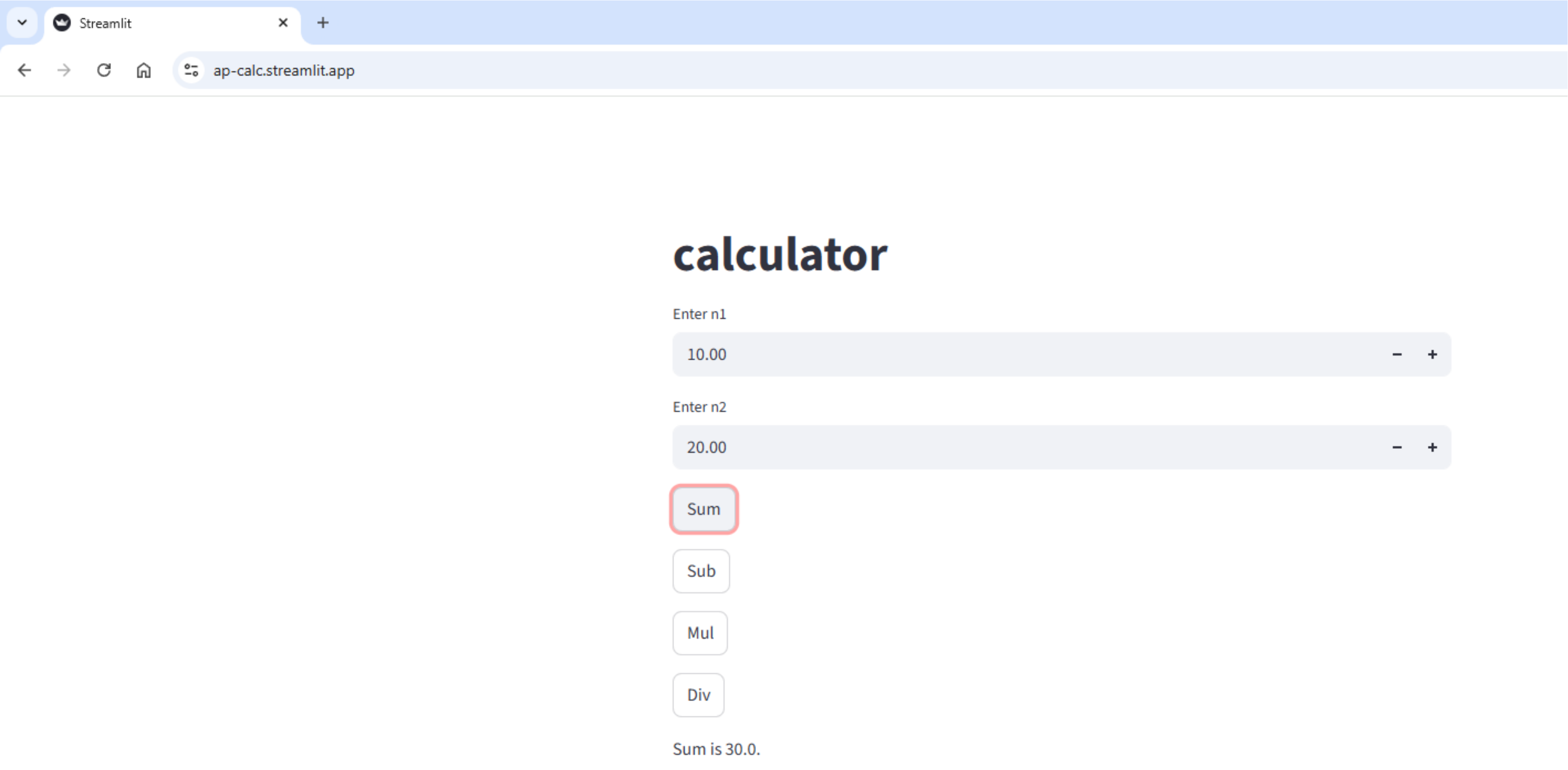
App URL (optional)

ap-calc .streamlit.app

Domain is available

Advanced settings

Deploy


Streamlit
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Enter n2
20.00
Sum
Sub
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Div
Sum is 30.0.

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Akash Padhiyar

Tech Educator • Consultant Tech Mentor • Founder

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